

A Counterexample to Köthe's Conjecture

Abstract

We construct a two-generated nil nonunital algebra N over $\overline{\mathbb{Q}}$ such that $M_2(N)$ contains a nonnilpotent element. It follows directly that there is a unital ring containing two nil left ideals whose sum is not nil.

1 Introduction

Köthe's conjecture asserts that the sum of two nil left ideals of a ring is nil [4]. We give a direct counterexample by constructing a two-generated nil algebra N for which $M_2(N)$ contains a nonnilpotent element. The construction is derived from the machine-checked formal counterexample in [1]. We use the minimum-rank argument and the reduction to a two-generated nil algebra from Adamczewski, Böhmeler, and Marczinzik [2, Lemma 4.6 and Theorem 1.1].

2 Statement and notation

An element x of a ring is *nilpotent* if $x^n = 0$ for some $n \geq 1$. A ring, algebra, or one-sided or two-sided ideal is *nil* if each of its elements is nilpotent.

All rings are associative. A *nonunital k -algebra* is a k -vector space equipped with an associative k -bilinear multiplication and is not assumed to have an identity. We write $M_d(B)$ for the ring of $d \times d$ matrices over a ring B , and $\mathbb{N} = \{0, 1, 2, \dots\}$. If B is a ring, $B[s]$ denotes the polynomial ring in a central indeterminate s . If $\tilde{B}$ is a unital k -algebra and $r_1, \dots, r_m \in \tilde{B}$, we write

$$k\langle r_1, \dots, r_m \rangle_+$$

for the nonunital k -subalgebra consisting of finite k -linear combinations of nonempty products of the r_i .

Theorem 2.1 (Main theorem). *There exists a nil nonunital $\overline{\mathbb{Q}}$ -algebra N , generated by two elements b, c , such that*

$$W = \begin{pmatrix} b^2 + c & bc \\ b & c \end{pmatrix} \in M_2(N)$$

is not nilpotent. Consequently there exists a unital ring containing two nil left ideals whose sum is not nil.

Until Section 7, k denotes a field. Algebraic closedness is imposed explicitly where needed.

3 A projective common-zero lemma

Let $h \geq 1$, and for each $1 \leq j \leq h$ let

$$y^{(j)} = (y_0^{(j)}, y_1^{(j)}, y_2^{(j)})$$

be a block of three variables. A polynomial in all these variables is *multihomogeneous of degree r in every block* if each monomial occurring with nonzero coefficient has total degree r in the variables belonging to each fixed block. The zero polynomial is regarded as multihomogeneous of every multidegree.

Lemma 3.1 (Projective common-zero lemma). *Let k be algebraically closed, let $h, r \geq 1$, and let*

$$f_1, \dots, f_q \in k[y_i^{(j)} : 1 \leq j \leq h, 0 \leq i \leq 2]$$

be multihomogeneous of degree r in every block. If $q \leq 2h$, then there exist

$$v^{(1)}, \dots, v^{(h)} \in k^3 \setminus \{0\}$$

such that

$$f_a(v^{(1)}, \dots, v^{(h)}) = 0 \quad (1 \leq a \leq q).$$

Proof. Set

$$X = (\mathbb{P}_k^2)^h, \quad \dim X = 2h.$$

Let $\mathcal{M}_r$ be the set of monomials of total degree r in three variables, so

$$|\mathcal{M}_r| = \binom{r+2}{2}.$$

The product of the degree- r Veronese embeddings followed by the Segre embedding gives a closed immersion

$$\nu : X \longrightarrow \mathbb{P}_k^{M-1}, \quad M = |\mathcal{M}_r|^h,$$

whose homogeneous coordinates are indexed by tuples $(m_1, \dots, m_h) \in \mathcal{M}_r^h$ and are

$$\prod_{j=1}^h m_j(y^{(j)}).$$

Thus every polynomial of multidegree $(r, \dots, r)$ is the pullback under ν of a homogeneous linear form on $\mathbb{P}^{M-1}$.

For each a let ℓ_a be the corresponding linear form, and let

$$L = V(\ell_1, \dots, \ell_q) \subseteq \mathbb{P}^{M-1}.$$

The common kernel of q linear forms in an M -dimensional vector space has dimension at least $M - q$. Since

$$M \geq 3^h > 2h \geq q,$$

this kernel is nonzero, and therefore L is a nonempty projective linear subspace satisfying

$$\dim L \geq M - 1 - q.$$

Consequently

$$\dim \nu(X) + \dim L \geq 2h + M - 1 - q \geq M - 1.$$

Both $\nu(X)$ and L are projective varieties. By the projective dimension theorem, $\nu(X) \cap L \neq \emptyset$; see [3, Chapter I, Theorem 7.2]. A point of X is represented by h nonzero triples in k^3 , and belonging to L says precisely that all the f_a vanish. This proves the lemma. $\square$

4 Finite linearization of a noncommutative polynomial

Let

$$k\langle X_0, X_1, X_2 \rangle_+ = k\langle X_0, X_1, X_2 \rangle_+$$

denote the free nonunital associative k -algebra on three generators. Thus its k -basis is the set of all nonempty words in X_0, X_1, X_2 .

Lemma 4.1 (Finite linearization). *For every $f \in k\langle X_0, X_1, X_2 \rangle_+$ there exist an integer $d \geq 1$ and matrices*

$$C_0, C_1, C_2, D_0, D_1, D_2 \in M_d(k),$$

depending only on f , such that each D_i has all rows zero except possibly its first row and the following holds. For every unital k -algebra $\tilde{B}$ and every $a_0, a_1, a_2 \in \tilde{B}$, put

$$T_f(a; s) = \sum_{i=0}^2 (C_i + sD_i)a_i \in M_d(\tilde{B}[s]), \quad (1)$$

where $(C_i a_i)_{pq} = (C_i)_{pq} a_i$ and similarly for $D_i a_i$. If $T_f(a; s)$ is nilpotent, then $f(a_0, a_1, a_2)$ is nilpotent in $\tilde{B}$.

Proof. If $f = 0$, take $d = 1$ and $C_i = D_i = 0$ for $i = 0, 1, 2$; the conclusion is immediate. We may therefore assume $f \neq 0$. Write f uniquely as a finite sum of distinct nonempty words,

$$f = \sum_{\alpha=1}^m c_\alpha X_{i_{\alpha,1}} X_{i_{\alpha,2}} \cdots X_{i_{\alpha,\ell_\alpha}}, \quad (2)$$

where $c_\alpha \in k \setminus \{0\}$ and $\ell_\alpha \geq 1$.

Introduce one distinguished index, denoted 0. For every word with $\ell_\alpha \geq 2$, introduce additional indices

$$(\alpha, 2), (\alpha, 3), \dots, (\alpha, \ell_\alpha).$$

Let Ω be the resulting finite index set; choose any ordering with the distinguished index first, and put $d = |\Omega|$.

Let $\tilde{B}$ and a_0, a_1, a_2 be arbitrary as in the statement. Define a matrix $T(s) \in M_d(\tilde{B}[s])$ by the following nonzero entries:

$$T_{0,0}(s) = s \sum_{\ell_\alpha=1} c_\alpha a_{i_{\alpha,1}}, \quad (3)$$

$$T_{0,(\alpha,2)}(s) = s c_\alpha a_{i_{\alpha,1}} \quad (\ell_\alpha \geq 2), \quad (4)$$

$$T_{(\alpha,j),(\alpha,j+1)}(s) = a_{i_{\alpha,j}} \quad (2 \leq j < \ell_\alpha), \quad (5)$$

$$T_{(\alpha,\ell_\alpha),0}(s) = a_{i_{\alpha,\ell_\alpha}} \quad (\ell_\alpha \geq 2). \quad (6)$$

The positions and scalar coefficients in these formulas determine matrices $C_i, D_i \in M_d(k)$ depending only on f , with each D_i zero outside the distinguished row, and $T(s) = T_f(a; s)$ for every choice of $\tilde{B}$ and a_0, a_1, a_2 . Thus these matrices C_i, D_i depend only on f , as required.

Suppose that $T(s)^N = 0$ for some $N \geq 1$. In the matrix ring over the unital ring $\tilde{B}[s]$ we have

$$(1 - T(s))Q(s) = 1, \quad Q(s) = 1 + T(s) + \cdots + T(s)^{N-1}.$$

Consider the column of $Q(s)$ indexed by 0. Write

$$q_0 = Q_{0,0}(s), \quad q_{\alpha,j} = Q_{(\alpha,j),0}(s).$$

The $(\alpha, j), 0$ entries of $(1 - T)Q = 1$ give, successively,

$$\begin{aligned} q_{\alpha, \ell_\alpha} &= a_{i_\alpha, \ell_\alpha} q_0, \\ q_{\alpha, j} &= a_{i_\alpha, j} q_{\alpha, j+1} \quad (2 \leq j < \ell_\alpha). \end{aligned}$$

Therefore

$$q_{\alpha, 2} = a_{i_\alpha, 2} a_{i_\alpha, 3} \cdots a_{i_\alpha, \ell_\alpha} q_0. \quad (7)$$

The $0, 0$ entry of $(1 - T)Q = 1$, together with (3), (4), and (7), is

$$q_0 = 1 + s f(a_0, a_1, a_2) q_0.$$

The constant coefficient of this identity is 1, so $q_0 \neq 0$. Write

$$q_0 = b_0 + b_1 s + \cdots + b_e s^e, \quad b_e \neq 0.$$

Since s is central, comparison of coefficients gives

$$b_0 = 1, \quad b_{n+1} = f(a_0, a_1, a_2) b_n \quad (0 \leq n < e),$$

and the coefficient of s^{e+1} gives

$$f(a_0, a_1, a_2) b_e = 0.$$

Inductively $b_n = f(a_0, a_1, a_2)^n$, and hence

$$f(a_0, a_1, a_2)^{e+1} = 0.$$

Thus $f(a_0, a_1, a_2)$ is nilpotent. □

For each nonzero $f \in k\langle X_0, X_1, X_2 \rangle_+$, choose matrices $C_i(f), D_i(f)$ furnished by Lemma 4.1. For a scalar triple $y = (y_0, y_1, y_2) \in k^3$ define

$$P_f(y; s) = \sum_{i=0}^2 y_i (C_i(f) + s D_i(f)) \in M_{d_f}(k[s]). \quad (8)$$

The variable s occurs only in the first row of this matrix.

5 A degree bound for minors

The restriction on the s -coefficient gives the following degree bound.

Lemma 5.1 (Degree and multihomogeneity of minors). *Let*

$$P(y; s) = \sum_{i=0}^2 y_i (C_i + s D_i) \in M_d(k[s]),$$

where every D_i is zero outside the first row. Consider a product of L matrices of this form. Some factors may be evaluated at fixed vectors of k^3 , and, for some $m \geq 0$, the remaining factors are evaluated at pairwise independent variable triples

$$y^{(1)}, \dots, y^{(m)}.$$

Assume that each variable triple occurs in exactly one factor, and fix an integer r with $1 \leq r \leq d$. Then every $r \times r$ minor $F(s)$ of the product satisfies:

- (i) $\deg_s F \leq L$ if $F \neq 0$;
- (ii) for every $h \geq 0$, the coefficient $[s^h]F$ is multihomogeneous of degree r in each variable triple $y^{(j)}$.

Proof. For one factor, every matrix entry is linear in the three coordinates of y , so every $r \times r$ minor is homogeneous of degree r in y . Moreover s occurs in at most one selected row. In the determinant expansion, therefore, each term contains at most one entry involving s , and the degree in s is at most one.

For a product, apply the Cauchy–Binet formula repeatedly. Every $r \times r$ minor of a product is a sum of products of $r \times r$ minors, one from each factor. Hence the s -degree of every summand is at most L . If $y^{(j)}$ occurs in one factor, the minor taken from that factor is homogeneous of degree r in $y^{(j)}$, while all other factors are independent of $y^{(j)}$. Thus every summand is multihomogeneous of degree r in each variable block. Summation, and then taking a coefficient of s , preserves this property. The zero polynomial causes no exception because it is multihomogeneous of every degree by convention. $\square$

6 A zero product with prescribed periodic values

Fix an integer $Q \geq 1$, a subset

$$S \subseteq \{0, 1, \dots, Q-1\},$$

and a nonzero vector $c_r \in k^3$ for each $r \in S$. At every nonnegative integer position congruent to r modulo Q , the value c_r is prescribed; no value is prescribed on the other residue classes. A finite word

$$w = (w_0, \dots, w_{\ell-1}) \in (k^3 \setminus \{0\})^\ell$$

is compatible with the prescribed values if

$$w_j = c_r \quad \text{whenever } j \equiv r \pmod{Q} \text{ and } r \in S.$$

Set

$$H = Q - |S|.$$

Thus H is the number of residue classes modulo Q on which no value has been prescribed. We shall use the condition

$$|S| < \frac{Q}{2}, \quad \text{equivalently} \quad Q < 2H. \quad (9)$$

Lemma 6.1 (Rank factorization from a nonsingular minor). *Let E be a field and let $A \in M_d(E)$ have rank $r > 0$. Choose sets I, J of r row and column indices such that*

$$M = A_{I,J}$$

is invertible. Put

$$U = A_{\bullet, J} \in M_{d \times r}(E), \quad V = A_{I, \bullet} \in M_{r \times d}(E).$$

Then

$$A = UM^{-1}V. \tag{10}$$

Consequently, for every $B \in M_d(E)$,

$$ABA = UM^{-1}(VBU)M^{-1}V, \tag{11}$$

and

$$(ABA)_{I, J} = VBU. \tag{12}$$

In particular, if $\det((ABA)_{I, J}) = 0$, then $\text{rank}(ABA) < r$.

Proof. The r columns of U are linearly independent because their restriction to the rows I is the invertible matrix M . Since $\text{rank } A = r$, they form a basis of the column space of A . Thus the j th column of A has the form $U\lambda_j$ for a unique $\lambda_j \in E^r$. Restricting to the rows I gives

$$V_{\bullet, j} = M\lambda_j,$$

so $\lambda_j = M^{-1}V_{\bullet, j}$. This proves (10), and (11) follows by substitution. Since $U_{I, \bullet} = M$ and $V_{\bullet, J} = M$, restricting (11) to rows I and columns J gives (12).

If $\det((ABA)_{I, J}) = 0$, then (12) gives $\text{rank}(VBU) < r$. Equation (11) therefore yields

$$\text{rank}(ABA) \leq \text{rank}(VBU) < r.$$

□

For a finite word $w = (w_0, \dots, w_{\ell-1})$, write

$$P[w] = P(w_0; s)P(w_1; s) \cdots P(w_{\ell-1}; s).$$

Proposition 6.2 (A zero product with prescribed periodic values). *Let k be algebraically closed and let*

$$P(y; s) = \sum_{i=0}^2 y_i(C_i + sD_i) \in M_d(k[s]),$$

where each D_i is zero outside the first row. Suppose that values have been prescribed on fewer than half of the residue classes modulo Q as above. Then there exists a compatible word w of positive length divisible by Q such that

$$P[w] = 0.$$

Proof. Let $H = Q - |S|$, so $Q < 2H$. Compatible words of positive length divisible by Q exist: for example, take a word of length Q and fill every unprescribed position with any fixed nonzero vector of k^3 .

Among all such compatible words, choose one, say w , for which

$$\text{rank}_{k(s)} P[w]$$

is minimal. This minimum exists because the rank belongs to the finite set $\{0, 1, \dots, d\}$. Write $\ell = |w|$ and

$$A = P[w].$$

If $\text{rank}_{k(s)} A = 0$, then every entry of A is zero in $k(s)$ and hence in $k[s]$, so $P[w] = 0$ and we are done.

Suppose instead that

$$r = \text{rank}_{k(s)} A > 0.$$

Choose row and column sets I, J of cardinality r such that $A_{I,J}$ is invertible over $k(s)$. Put

$$m = 2\ell + 1.$$

Between two copies of w insert a block c of length mQ . At positions belonging to prescribed residue classes, use the prescribed vectors. At every other position introduce a separate triple of variables. There are

$$N_0 = mH$$

such variable triples.

Consider

$$F(s) = \det(P[w c w]_{I,J}).$$

By Lemma 5.1, F has degree at most $2\ell + mQ$ in s if it is nonzero, and each coefficient of F is multihomogeneous of degree r in each of the N_0 variable triples. Thus the condition $F = 0$ is imposed by at most

$$2\ell + mQ + 1 = m(Q + 1)$$

multihomogeneous equations. Since $Q < 2H$ and Q, H are integers,

$$Q + 1 \leq 2H,$$

and therefore

$$2\ell + mQ + 1 = m(Q + 1) \leq 2mH = 2N_0.$$

The projective common-zero lemma gives nonzero values for all variable triples such that every coefficient of F vanishes. After this specialization, c is a word of nonzero vectors compatible with the prescribed values and

$$\det(P[w c w]_{I,J}) = 0. \tag{13}$$

Put $B = P[c]$ and regard A, B as matrices over $k(s)$. Since

$$P[w c w] = ABA,$$

Lemma 6.1 and (13) imply

$$\text{rank}_{k(s)} P[w c w] < r.$$

The word $w c w$ is compatible: the lengths of both w and c are divisible by Q , so the residue classes are unchanged under concatenation. Its length is positive and divisible by Q . This contradicts the minimal choice of w .

Hence the minimum rank is zero, and a compatible word with zero product exists. $\square$

7 A sequence with vanishing consecutive products

For the remainder of the proof, fix an algebraic closure $k = \overline{\mathbb{Q}}$. It is algebraically closed by definition and countable because $\mathbb{Q}[x]$ is countable and each nonzero polynomial has only finitely many roots in k . The set of nonempty words in three letters is countable, so $k\langle X_0, X_1, X_2 \rangle_+$ has a countable k -basis; since elements of $k\langle X_0, X_1, X_2 \rangle_+$ are finite k -linear combinations of basis words, $k\langle X_0, X_1, X_2 \rangle_+$ itself is countable. Enumerate the nonzero elements of $k\langle X_0, X_1, X_2 \rangle_+$ as

$$f_1, f_2, f_3, \dots,$$

and let $P_j(y; s) = P_{f_j}(y; s)$ be the matrix defined in (8).

Proposition 7.1 (A sequence with vanishing consecutive products). *There exists a sequence*

$$v_0, v_1, v_2, \dots \in k^3 \setminus \{0\}$$

with the following property: for every j there exists an integer $N_j \geq 1$ such that for every $n \geq 0$,

$$P_j(v_n; s)P_j(v_{n+1}; s) \cdots P_j(v_{n+N_j-1}; s) = 0. \quad (14)$$

Proof. Inductively prescribe nonzero vectors on residue classes modulo periods Q_j . At every stage, fewer than half of the residue classes have prescribed values.

At stage 0, take period $Q_0 = 1$ and assign nothing.

Suppose stage $j - 1$ has period Q_{j-1} , with values prescribed on a_{j-1} residue classes and

$$\frac{a_{j-1}}{Q_{j-1}} < \frac{1}{2}.$$

Apply Proposition 6.2 to P_j and the values prescribed at stage $j - 1$. We obtain a word compatible with those values

$$w^{(j)} = (w_0^{(j)}, \dots, w_{\ell_j-1}^{(j)})$$

of nonzero vectors, with

$$Q_{j-1} \mid \ell_j$$

and

$$P_j(w_0^{(j)}; s) \cdots P_j(w_{\ell_j-1}^{(j)}; s) = 0. \quad (15)$$

Choose an integer $M_j \geq 1$ so large that, with

$$Q_j = M_j Q_{j-1},$$

$$\frac{a_{j-1}}{Q_{j-1}} + \frac{\ell_j}{Q_j} < \frac{1}{2}. \quad (16)$$

This is possible because the first term on the left is already strictly less than $1/2$. In particular, $\ell_j/Q_j < 1/2$, so $\ell_j < Q_j$.

Define the prescribed values modulo Q_j as follows. First copy every previously prescribed value to all of its lifts modulo Q_j : a residue class modulo Q_j receives that value whenever its reduction modulo Q_{j-1} was assigned. Then prescribe

$$w_0^{(j)}, \dots, w_{\ell_j-1}^{(j)}$$

at the first ℓ_j residue classes modulo Q_j . There is no conflict because $w^{(j)}$ respects all values prescribed at stage $j - 1$.

The lifted previously prescribed values occupy exactly $M_j a_{j-1}$ residue classes. Adding the first ℓ_j classes creates at most ℓ_j additional assignments. Hence the new number a_j of assigned classes satisfies

$$a_j \leq M_j a_{j-1} + \ell_j.$$

Dividing by $Q_j = M_j Q_{j-1}$ and using (16) gives

$$\frac{a_j}{Q_j} < \frac{1}{2}.$$

Thus the induction continues, and every value prescribed at one stage is retained at every later stage.

Define the full sequence (v_n) as follows. If the position n is prescribed at some stage, let v_n be that prescribed value; this is well defined because later stages preserve it. If n is never prescribed, set

$$v_n = (1, 0, 0).$$

Every v_n is nonzero, and the full sequence agrees with every value prescribed at every stage.

Fix j . At stage j , the word $w^{(j)}$ occupies the first ℓ_j positions of every period Q_j , and $\ell_j < Q_j$. Hence the final sequence contains $w^{(j)}$ beginning at every multiple of Q_j .

Let $n \geq 0$, and let b be the least multiple of Q_j satisfying $b \geq n$. Then

$$0 \leq b - n < Q_j.$$

Since $\ell_j < Q_j$, the interval of indices

$$b, b + 1, \dots, b + \ell_j - 1$$

is contained in

$$n, n + 1, \dots, n + 2Q_j - 1.$$

Therefore the matrix product over the latter interval contains, as a contiguous subproduct, the zero product (15). By associativity and absorption by zero,

$$P_j(v_n; s) \cdots P_j(v_{n+2Q_j-1}; s) = 0.$$

Thus (14) holds with $N_j = 2Q_j$. □

8 Weighted backward shifts

Let (v_n) be the sequence from Proposition 7.1. Put

$$K = k(t), \quad V = K^{\mathbb{N}},$$

the K -vector space of all sequences $u = (u(0), u(1), \dots)$ with entries in K . Multiplication in $\text{End}_K(V)$ is composition: $(fg)(u) = f(g(u))$. For $i = 0, 1, 2$ define a K -linear endomorphism $a_i \in \text{End}_K(V)$ by

$$(a_i u)(n) = v_n(i) u(n + 1). \tag{17}$$

Let

$$A = k\langle a_0, a_1, a_2 \rangle_+ \subseteq \text{End}_K(V) \quad (18)$$

be the nonunital k -subalgebra generated by these three endomorphisms.

We first connect products of the scalar matrices $P(v_n; s)$ with powers of the corresponding matrix of endomorphisms.

Let

$$P(y; s) = \sum_{i=0}^2 y_i (C_i + sD_i) \in M_d(k[s])$$

be any matrix of the form used above. Define

$$\hat{P}(a; s) = \sum_{i=0}^2 (C_i + sD_i) a_i \in M_d(\text{End}_K(V)[s]). \quad (19)$$

Explicitly, its (p, q) entry is

$$\sum_{i=0}^2 (C_i)_{pq} a_i + s \sum_{i=0}^2 (D_i)_{pq} a_i.$$

Lemma 8.1 (Coefficient action of a power). *For every $N \geq 1$, every $h \geq 0$, every $1 \leq p, q \leq d$, every $u \in V$, and every $n \geq 0$,*

$$\left([s^h] (\hat{P}(a; s)^N)_{pq} u \right) (n) = \left([s^h] (P(v_n; s) \cdots P(v_{n+N-1}; s))_{pq} \right) u(n + N). \quad (20)$$

On the right, the coefficient belongs to k and is viewed inside K .

Proof. We argue by induction on N .

For $N = 1$, the only possibly nonzero coefficients are those of s^0 and s^1 . By (17),

$$\begin{aligned} \left(\sum_i (C_i)_{pq} a_i \right) u(n) &= \left(\sum_i v_n(i) (C_i)_{pq} \right) u(n + 1), \\ \left(\sum_i (D_i)_{pq} a_i \right) u(n) &= \left(\sum_i v_n(i) (D_i)_{pq} \right) u(n + 1), \end{aligned}$$

which is precisely (20).

Assume the formula holds for N . Since

$$\hat{P}^{N+1} = \hat{P} \hat{P}^N,$$

the coefficient of s^h in its (p, q) entry is

$$\sum_{r=1}^d \sum_{a+b=h} [s^a] \hat{P}_{pr} [s^b] (\hat{P}^N)_{rq},$$

where multiplication in $\text{End}_K(V)$ is composition. Apply this endomorphism to u and evaluate at n . Since multiplication in $\text{End}_K(V)$ is composition, the right factor acts on u first; the left factor then evaluates that result at $n + 1$. Using the case $N = 1$ for the left factor and the induction hypothesis for the right factor gives

$$([s^a] P(v_n; s)_{pr}) \left([s^b] (P(v_{n+1}; s) \cdots P(v_{n+N}; s))_{rq} \right) u(n + N + 1).$$

Summing over the intermediate matrix index r and over $a + b = h$ is exactly the coefficient rule for the matrix product

$$P(v_n; s)P(v_{n+1}; s) \cdots P(v_{n+N}; s).$$

This proves (20) for $N + 1$. □

Corollary 8.2. *If for some $N \geq 1$*

$$P(v_n; s) \cdots P(v_{n+N-1}; s) = 0 \quad \text{for every } n \geq 0,$$

then

$$\hat{P}(a; s)^N = 0 \quad \text{in } M_d(\text{End}_K(V)[s]).$$

Proof. Every coefficient of every entry of $\hat{P}(a; s)^N$ acts as zero on every $u \in V$ at every coordinate n by Lemma 8.1. Hence each coefficient is the zero endomorphism, so the polynomial matrix itself is zero. □

9 The algebra generated by the shifts is nil

Proposition 9.1. *The algebra A defined in (18) is nil.*

Proof. Let $x \in A$. If $x = 0$, there is nothing to prove. Assume $x \neq 0$. By definition of the nonunital algebra generated by a_0, a_1, a_2 , there exists a nonzero $f \in k\langle X_0, X_1, X_2 \rangle_+$ such that

$$x = f(a_0, a_1, a_2).$$

Choose j with $f_j = f$ in the enumeration of Section 7. Let $P_j = P_f$ be its scalar linearization matrix.

By Proposition 7.1, there is $N_j \geq 1$ such that every consecutive product of length N_j of $P_j(v_n; s)$ is zero. Corollary 8.2 therefore shows that the corresponding matrix of endomorphisms

$$T_f(a; s) = \hat{P}_j(a; s)$$

is nilpotent. Lemma 4.1 now implies that

$$f(a_0, a_1, a_2) = x$$

is nilpotent. Since x was arbitrary, A is nil. □

10 A two-generated nil algebra with a nonnilpotent matrix

For each $n \geq 0$, define

$$p_n(t) = v_n(0) + t v_n(1) + t^2 v_n(2) \in k[t]. \tag{21}$$

Because $v_n \neq 0$, the polynomial $p_n(t)$ is nonzero and hence invertible in $K = k(t)$.

Define $z \in V$ by

$$z(n) = \left(\prod_{j=0}^{n-1} p_j(t) \right)^{-1}, \tag{22}$$

with the empty product equal to 1. Then $z(0) = 1$, so $z \neq 0$, and

$$p_n(t)z(n+1) = z(n).$$

By the definition of the weighted shifts,

$$(a_0 + ta_1 + t^2a_2)z = z. \quad (23)$$

Since A is nil and $a_0 \in A$, choose $m \geq 1$ with $a_0^m = 0$. In the unital ring $\text{End}_K(V)$ put

$$g = 1 + a_0 + a_0^2 + \cdots + a_0^{m-1}.$$

Then

$$g(1 - a_0) = 1 - a_0^m = 1. \quad (24)$$

Set

$$b = ga_1, \quad c = ga_2. \quad (25)$$

Both b and c belong to A , because

$$b = a_1 + a_0a_1 + \cdots + a_0^{m-1}a_1, \quad c = a_2 + a_0a_2 + \cdots + a_0^{m-1}a_2.$$

Define

$$N = k\langle b, c \rangle_+ \subseteq A.$$

Since A is nil, every element of its subalgebra N is nilpotent. Thus N is a two-generated nil k -algebra.

Rearranging (23) gives

$$(1 - a_0)z = ta_1z + t^2a_2z.$$

Apply g and use (24). Since g is K -linear,

$$z = tbz + t^2cz,$$

so

$$bz + tcz = t^{-1}z. \quad (26)$$

Consider the K -linear endomorphism of $V \oplus V$ represented by

$$H = \begin{pmatrix} b & c \\ 1 & 0 \end{pmatrix}, \quad \xi = \begin{pmatrix} z \\ tz \end{pmatrix}.$$

Equation (26) gives

$$H\xi = \begin{pmatrix} bz + tcz \\ z \end{pmatrix} = \begin{pmatrix} t^{-1}z \\ z \end{pmatrix} = t^{-1}\xi.$$

Therefore $H^2\xi = t^{-2}\xi$. Direct multiplication gives

$$W := H^2 = \begin{pmatrix} b^2 + c & bc \\ b & c \end{pmatrix}. \quad (27)$$

Every entry of W lies in N , hence $W \in M_2(N)$. Since $\xi \neq 0$, for every $r \geq 1$,

$$W^r\xi = t^{-2r}\xi \neq 0.$$

Thus W is not nilpotent. This proves the first assertion of Theorem 2.1.

11 Two nil left ideals whose sum is not nil

Set

$$R = k1 + N \subseteq \text{End}_K(V).$$

This is a unital k -algebra: for $\lambda, \mu \in k$ and $x, y \in N$,

$$(\lambda 1 + x)(\mu 1 + y) = \lambda\mu 1 + (\lambda y + \mu x + xy) \in R.$$

Moreover N is a two-sided ideal of R , because

$$(\lambda 1 + x)y = \lambda y + xy \in N, \quad y(\lambda 1 + x) = \lambda y + yx \in N.$$

Let

$$S = M_2(R)$$

and define

$$L_1 = \left\{ \begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix} : x, y \in N \right\}, \quad L_2 = \left\{ \begin{pmatrix} 0 & x \\ 0 & y \end{pmatrix} : x, y \in N \right\}.$$

Since N is an additive subgroup of R , both L_1 and L_2 are additive subgroups of S . For

$$M = \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix} \in S$$

and $x, y \in N$, one has

$$M \begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix} = \begin{pmatrix} r_{11}x + r_{12}y & 0 \\ r_{21}x + r_{22}y & 0 \end{pmatrix} \in L_1$$

and

$$M \begin{pmatrix} 0 & x \\ 0 & y \end{pmatrix} = \begin{pmatrix} 0 & r_{11}x + r_{12}y \\ 0 & r_{21}x + r_{22}y \end{pmatrix} \in L_2,$$

because N is a left ideal of R . Thus L_1 and L_2 are left ideals of S .

They are nil. If

$$X = \begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix} \in L_1,$$

then

$$X^n = \begin{pmatrix} x^n & 0 \\ yx^{n-1} & 0 \end{pmatrix} \quad (n \geq 1).$$

Since x is nilpotent, so is X . Similarly, if

$$Y = \begin{pmatrix} 0 & x \\ 0 & y \end{pmatrix} \in L_2,$$

then

$$Y^n = \begin{pmatrix} 0 & xy^{n-1} \\ 0 & y^n \end{pmatrix} \quad (n \geq 1),$$

so Y is nilpotent because y is nilpotent.

Finally,

$$L_1 + L_2 = M_2(N),$$

and this sum contains the nonnilpotent matrix W from (27). Thus L_1 and L_2 are nil left ideals of the unital ring S , whereas $L_1 + L_2$ is not nil. This completes the proof of Theorem 2.1.

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